The Classical Kicked Top

Lecture notes on the physics and on the `classical_kicked_top` code

Abstract

These notes explain the classical kicked top from the ground up: angular momentum as a dynamical variable, the sphere as a phase space, the kicked-top Hamiltonian and its stroboscopic map, energy, symmetries, fixed points and their stability, the transition from regular to chaotic motion, and Lyapunov exponents, with a short outlook on the quantum kicked top. The second part describes the numerical method and the structure of the Fortran/Python code, pointing each piece of code to the equation it implements. The only prerequisites are Hamiltonian mechanics (Hamilton's equations, Poisson brackets) and basic linear algebra. All phase-space figures were produced with the code described in part II.

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Part I
Physics

1 Angular momentum as a dynamical variable

1.1 Poisson brackets of angular momentum

Recall that for canonical coordinates q_a and momenta p_a the Poisson bracket of two phase-space functions is

$$\{A, B\} = \sum_a \left(\frac{\partial A}{\partial q_a}\frac{\partial B}{\partial p_a} - \frac{\partial A}{\partial p_a}\frac{\partial B}{\partial q_a} \right), \qquad \{q_a, p_b\} = \delta_{ab}, \tag{1}$$

and that any observable evolves as

$$\frac{dA}{dt} = \{A, H\} + \frac{\partial A}{\partial t}. \tag{2}$$

Take a particle with position $\boldsymbol{r}$ and momentum $\boldsymbol{p}$ and its angular momentum $\boldsymbol{J} = \boldsymbol{r} \times \boldsymbol{p}$, i.e. $J_i = \epsilon_{ijk} r_j p_k$ (sum over repeated indices; ϵ_{ijk} is the Levi-Civita symbol). A short calculation gives, for example,

$$\{J_x, J_y\} = \{yp_z - zp_y, zp_x - xp_z\} = y\{p_z, z\}p_x + p_y\{z, p_z\}x = -yp_x + xp_y = J_z,$$

and in general

$$\boxed{\{J_i, J_j\} = \epsilon_{ijk}\, J_k.} \tag{3}$$

The key idea of these notes is that (3) is *all* we need. We can forget $\boldsymbol{r}$ and $\boldsymbol{p}$ and treat the three components J_x, J_y, J_z themselves as the dynamical variables, with the bracket (3). This is how one describes the orientation of a rigid body's angular momentum, the total angular momentum of many atoms, or the classical limit of a *spin*. A spin is an angular momentum that has no $\boldsymbol{r} \times \boldsymbol{p}$ behind it, but whose quantum operators satisfy $[\hat{J}_i, \hat{J}_j] = i\hbar\,\epsilon_{ijk}\hat{J}_k$, the quantum version of (3) (the usual correspondence is $\{\cdot,\cdot\} \leftrightarrow [\cdot,\cdot]/i\hbar$).

For any function $f(\boldsymbol{J})$, the chain rule and (3) give

$$\{J_i, f\} = \frac{\partial f}{\partial J_m}\{J_i, J_m\} = \epsilon_{imk}\,\frac{\partial f}{\partial J_m}\,J_k. \tag{4}$$

1.2 The length of $\boldsymbol{J}$ is conserved: the phase space is a sphere

Let $J^2 = J_x^2 + J_y^2 + J_z^2$. Using (4),

$$\{J_i, J^2\} = \epsilon_{imk}\, 2J_m J_k = 0, \tag{5}$$

because ϵ_{imk} is antisymmetric in m, k while $J_m J_k$ is symmetric. Hence J^2 has a vanishing bracket with every function of $\boldsymbol{J}$ (it is a *Casimir* of the bracket), and in particular with every Hamiltonian $H(\boldsymbol{J}, t)$:

$$\frac{dJ^2}{dt} = \{J^2, H\} = 0. \tag{6}$$

Whatever the Hamiltonian, the vector $\boldsymbol{J}$ moves on a sphere of fixed radius, which we call j:

$$|\boldsymbol{J}| = j. \tag{7}$$

This sphere is the phase space of the system. It is two-dimensional, so the system has *one degree of freedom*, just like a particle on a line. The difference is that the phase space is compact (finite area) and curved.

1.3 Canonical coordinates on the sphere

Parametrise the sphere with the usual polar angle θ (measured from the z axis) and azimuth ϕ (fig. 1):

$$\boldsymbol{J} = j\,(\sin\theta\cos\phi,\ \sin\theta\sin\phi,\ \cos\theta). \tag{8}$$

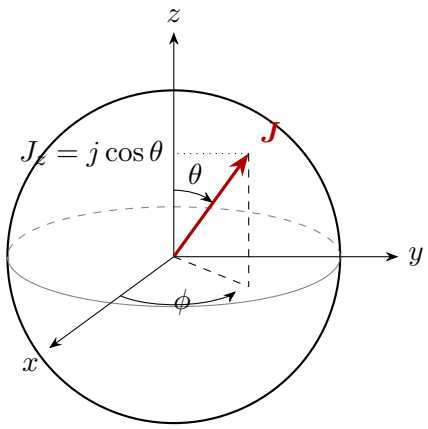


Figure 1: Coordinates on the phase-space sphere of radius j. The canonical pair is (ϕ, J_z).

Theorem. *The pair $(q,p) = (\phi, J_z)$ is canonical: $\{\phi, J_z\} = 1$.*

Proof. Write $J_x = R\cos\phi$, $J_y = R\sin\phi$ with $R = \sqrt{j^2 - J_z^2}$, so that $\partial R/\partial J_z = -J_z/R$, and compute brackets with $\{A,B\} = \partial_\phi A\,\partial_{J_z} B - \partial_{J_z} A\,\partial_\phi B$:

$$\{J_x, J_y\} = (-R\sin\phi)\Big(-\tfrac{J_z}{R}\sin\phi\Big) - \Big(-\tfrac{J_z}{R}\cos\phi\Big)(R\cos\phi) = J_z,$$
$$\{J_y, J_z\} = R\cos\phi = J_x, \qquad \{J_z, J_x\} = -(-R\sin\phi) = J_y.$$

This reproduces (3), so (ϕ, J_z) is a valid canonical pair. □

The phase-space (Liouville) area element is therefore $d\phi\,dJ_z$. The geometric area element of the sphere is $j^2\sin\theta\,d\theta\,d\phi$, and since $dJ_z = -j\sin\theta\,d\theta$,

$$j^2\sin\theta\,d\theta\,d\phi = j\ d\phi\,dJ_z. \tag{9}$$

Phase-space area and geometric area on the sphere are the same thing (up to the constant j). This is Archimedes' "hat-box" theorem: the band between two heights $z_1 < z_2$ has area proportional to $z_2 - z_1$. The total phase-space area is $4\pi j$. For this reason all phase-space plots in these notes use the axes $(\phi, \cos\theta)$ and not (ϕ,θ): equal areas in the plot are equal areas on the sphere.

1.4 Linear Hamiltonians generate rotations

Take $H = \boldsymbol{\omega}\cdot\boldsymbol{J}$ with a constant vector $\boldsymbol{\omega}$. With (2) and (4),

$$\frac{dJ_i}{dt} = \{J_i, \omega_m J_m\} = \epsilon_{imk}\,\omega_m J_k \quad\Longleftrightarrow\quad \frac{d\boldsymbol{J}}{dt} = \boldsymbol{\omega}\times\boldsymbol{J}. \tag{10}$$

This is *precession*. $\boldsymbol{J}$ rotates rigidly about the axis $\hat{\boldsymbol{\omega}}$ with angular velocity $|\boldsymbol{\omega}|$, counter-clockwise when looking down the axis (right-hand rule). After a time t the solution is $\boldsymbol{J}(t) = R_{\hat{\omega}}(|\boldsymbol{\omega}|t)\,\boldsymbol{J}(0)$. We will need the rotation matrices about y and about z:

$$R_y(\alpha) = \begin{pmatrix} \cos\alpha & 0 & \sin\alpha \\ 0 & 1 & 0 \\ -\sin\alpha & 0 & \cos\alpha \end{pmatrix}, \qquad R_z(\alpha) = \begin{pmatrix} \cos\alpha & -\sin\alpha & 0 \\ \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{11}$$

(Check: for $\boldsymbol{\omega} = \hat{\boldsymbol{y}}$, $\hat{\boldsymbol{y}} \times \boldsymbol{J} = (J_z, 0, -J_x)$, which is $\frac{d}{d\alpha} R_y(\alpha)\boldsymbol{J}$ at $\alpha = 0$.)

A Hamiltonian that is a function of $\boldsymbol{J}$ but not linear (for example J_z^2) still generates a rotation at each instant, but the axis and the rate depend on where $\boldsymbol{J}$ is. That nonlinearity is the source of chaos.

2 The kicked top

2.1 Hamiltonian

The kicked top is an angular momentum $\boldsymbol{J}$ with $|\boldsymbol{J}| = j$ and the time-periodic Hamiltonian

$$\boxed{H(t) = p\, J_y + \frac{k}{2j}\, J_z^2 \sum_{n=-\infty}^{\infty} \delta(t-n),} \tag{12}$$

with two real parameters:

- p, the angle of free precession about the y axis accumulated in one period (the period is set to 1);
- k, the strength of the *kick*, also called the *torsion* or *twist*.

The first term is linear, so it generates a uniform rotation about y (section 1.4). The second term acts only at the instants $t = n$ (integer n). It is quadratic in J_z: as shown below, it rotates $\boldsymbol{J}$ about the z axis by an angle proportional to J_z. Points on the upper hemisphere are turned one way, points on the lower hemisphere the other way, and the equator not at all, like twisting a towel. The factor $1/j$ makes the twist angle depend only on the *direction* of $\boldsymbol{J}$ (see below).

Because H depends explicitly on time, $dH/dt = \partial H/\partial t \neq 0$, so energy is *not* conserved (section 4). With one degree of freedom and no conserved quantity, the motion can be chaotic. In contrast, a time-independent one-degree-of-freedom system is always integrable.

2.2 Scaled variables: only the direction matters

Introduce the unit vector

$$\boldsymbol{s} = \frac{\boldsymbol{J}}{j} = (X, Y, Z) = (\sin\theta\cos\phi,\ \sin\theta\sin\phi,\ \cos\theta), \qquad |\boldsymbol{s}| = 1, \tag{13}$$

and write $H = j\,h$ with

$$h(\boldsymbol{s}, t) = p\, Y + \frac{k}{2}\, Z^2 \sum_n \delta(t-n). \tag{14}$$

From (4), $\frac{ds_i}{dt} = \frac{1}{j}\{J_i, H\} = \epsilon_{imk}\, \frac{\partial h}{\partial s_m}\, s_k$, i.e.

$$\boxed{\frac{d\boldsymbol{s}}{dt} = \nabla_s\, h \times \boldsymbol{s}.} \tag{15}$$

The radius j has dropped out. The classical dynamics of the direction $\boldsymbol{s}$ is the same for every j, so we can work on the unit sphere. The value of j only matters in the quantum theory, where $1/j$ plays the role of Planck's constant (section 9).

3 The stroboscopic map

Because the Hamiltonian is periodic and the kicks are instantaneous, it is natural to look at the system once per period, just after each kick. Let $\boldsymbol{s}_n$ be the state at $t = n^+$ (just after the kick at $t = n$). Integrating (15) from $t = n^+$ to $t = (n+1)^+$ gives a map $\boldsymbol{s}_{n+1} = F(\boldsymbol{s}_n)$, the *stroboscopic* or *Floquet* map. It is derived in two steps.

Step 1: free precession. For $n < t < n+1$ only the first term of (14) is present: $\nabla_s h = p\,\hat{\boldsymbol{y}}$, so $\dot{\boldsymbol{s}} = p\,\hat{\boldsymbol{y}} \times \boldsymbol{s}$. During one unit of time this is a rotation about y by the angle p, $\boldsymbol{s}' = R_y(p)\,\boldsymbol{s}_n$:

$$X' = X\cos p + Z\sin p, \qquad Y' = Y, \qquad Z' = -X\sin p + Z\cos p. \tag{16}$$

Step 2: the kick. Near $t = n+1$ the δ-function dominates and the finite precession term contributes nothing over the infinitesimal kick duration. Now $\nabla_s h = kZ\,\delta(t-n-1)\,\hat{\boldsymbol{z}}$ and

$$\dot{\boldsymbol{s}} = kZ\,\delta(t-n-1)\,\hat{\boldsymbol{z}} \times \boldsymbol{s}. \tag{17}$$

The z component of $\hat{\boldsymbol{z}} \times \boldsymbol{s}$ is zero, so Z does not change during the kick. The equation is then a rotation about z with angular velocity $kZ\,\delta(t-n-1)$, and integrating over the kick gives a rotation by the finite angle kZ (with $Z = Z'$ the value at the moment of the kick):

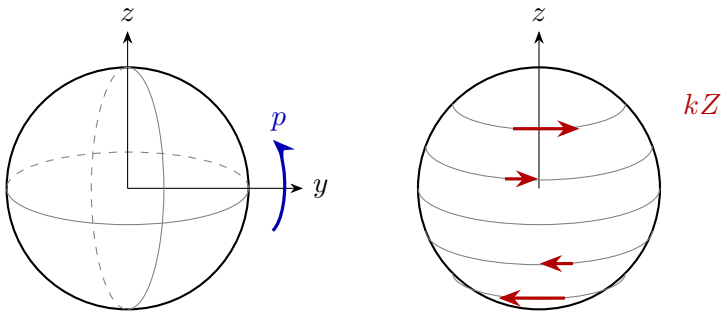
$$X'' = X'\cos(kZ') - Y'\sin(kZ'), \qquad Y'' = X'\sin(kZ') + Y'\cos(kZ'), \qquad Z'' = Z'. \tag{18}$$

This is a *torsion*: a rotation about z whose angle depends on the height Z.

The map. Together:

$$\boxed{\boldsymbol{s}_{n+1} = F(\boldsymbol{s}_n) = R_z\big(kZ'\big)\,R_y(p)\,\boldsymbol{s}_n, \qquad Z' = \big(R_y(p)\boldsymbol{s}_n\big)_z.} \tag{19}$$

Figure 2 illustrates the two steps. Since Z is unchanged by the torsion, the inverse map is simply $F^{-1}(\boldsymbol{s}) = R_y(-p)\,R_z(-kZ)\,\boldsymbol{s}$. Both steps are rotations, so $|\boldsymbol{s}| = 1$ is preserved exactly.



(a) rotation about y by p (b) torsion about z by kZ

Figure 2: The two steps of one period of the kicked top, eq. (19). (a) A rigid rotation about y. (b) A rotation about z whose angle kZ grows with height: the upper and lower hemispheres are twisted in opposite directions and the equator stays put.

3.1 The map preserves phase-space area

In the canonical coordinates (ϕ, Z) of section 1.3, the torsion (18) reads

$$\phi'' = \phi' + kZ', \qquad Z'' = Z', \tag{20}$$

whose Jacobian matrix $\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$ has determinant 1. This is the classic *twist map*. The rotation (16) is a Hamiltonian flow and therefore a canonical transformation. By Liouville's theorem it also preserves area. A composition of area-preserving maps is area-preserving, so

$$\det DF = 1 \quad \text{(in the coordinates } (\phi, Z)\text{)}. \tag{21}$$

Two consequences will be used repeatedly. First, there are no attractors: a region of phase space can be stretched and folded but never shrunk. Second, linear stability is decided by a single number, the trace of the Jacobian (section 6).

4 Energy

The code evaluates, at every recorded point $\boldsymbol{s}_n$, the two terms of the scaled Hamiltonian (14) without the δ comb, and their sum:

$$E_{\text{rot}} = pY, \qquad E_{\text{kick}} = \frac{k}{2}Z^2, \qquad E = E_{\text{rot}} + E_{\text{kick}} = pY + \frac{k}{2}Z^2. \tag{22}$$

E is *not* a constant of motion. The Hamiltonian depends on time, and each kick changes Y while each precession changes Z. There are, however, two reasons why E is a useful quantity.

The near-continuous limit. Suppose both p and k are small. Each period then moves $\boldsymbol{s}$ only a little, and the composition of the two small rotations is, to first order, the same as applying them simultaneously. One period is then approximately the time-1 flow of the *effective Hamiltonian*

$$h_{\text{eff}} = pY + \frac{k}{2}Z^2 = E, \tag{23}$$

with corrections of order pk (this is the first term of the Baker–Campbell–Hausdorff formula). A time-independent one-degree-of-freedom Hamiltonian is conserved, so in this limit E is nearly conserved and orbits follow the level curves of E. Figure 3 shows this for $p = 0.1$, $k = 0.2$: every orbit traces a curve of (almost) constant colour.

The stationary points of h_{eff} on the sphere (Lagrange condition $\nabla h_{\text{eff}} = \lambda \boldsymbol{s}$, i.e. $(0, p, kZ) = \lambda(X, Y, Z)$) are the poles $\pm\hat{\boldsymbol{y}}$ and, if $k > p$, the two points $(0,\ p/k,\ \pm\sqrt{1 - p^2/k^2})$. For $p = 0.1$, $k = 0.2$ these are at $\phi = \pi/2$, $Z = \pm 0.87$: the two centres visible in fig. 3. The pole $+\hat{\boldsymbol{y}}$ ($\phi = \pi/2$, $Z = 0$) has become a saddle, and the separatrix through it is clearly visible.

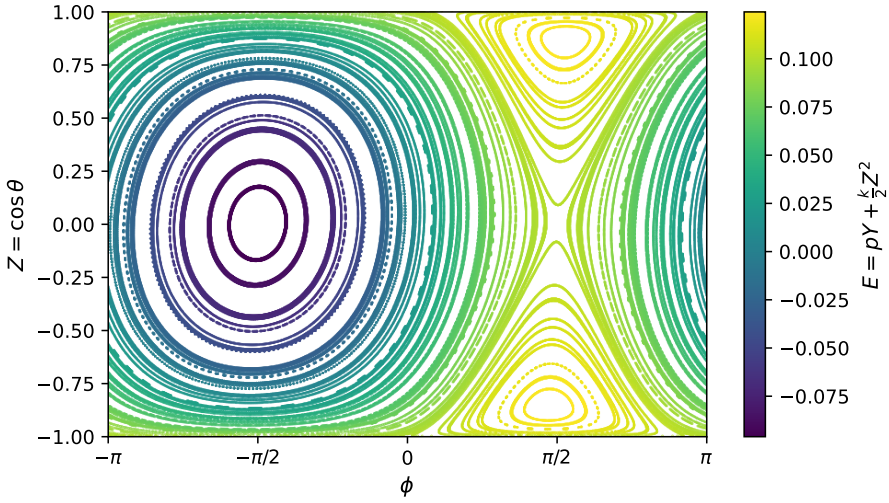


Figure 3: Phase portrait for small parameters $p = 0.1$, $k = 0.2$ (12×12 grid of initial conditions, 1000 kicks each), coloured by E of eq. (22). Orbits lie on level curves of E, as predicted by eq. (23).

Away from that limit. For $p = \pi/2$ the near-continuous picture does not apply and E fluctuates from kick to kick (fig. 4). On a regular orbit (left) the fluctuations are quasi-periodic and stay inside a narrow band. On a chaotic orbit (right) they are erratic, and E wanders over a large part of its allowed range $[-p,\ p + k/2]$. The behaviour of E_n is thus a simple diagnostic of regular versus chaotic motion.

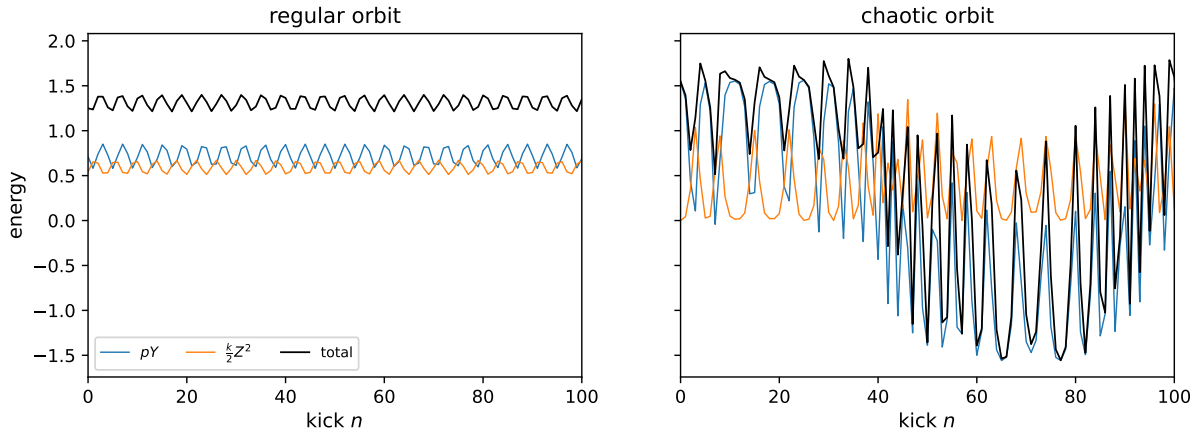


Figure 4: The energy terms (22) along a regular orbit (left, started inside the island around the fixed point $\boldsymbol{s}_+$ of section 6.3) and along a chaotic orbit (right, started next to the unstable pole $+\hat{\boldsymbol{y}}$), for $p = \pi/2$, $k = 3$.

5 Integrable limits and symmetry

Integrable limits. If $k = 0$, the map is a pure rotation $R_y(p)$ and Y is conserved. Every orbit lies on a circle $Y = \text{const}$. If $p/2\pi$ is rational, every orbit is periodic (for $p = \pi/2$, period 4). If $p = 0$, the map is the pure twist (20) and Z is conserved. Each circle of latitude rotates rigidly by its own angle kZ. In both cases the motion is *integrable*: there is a conserved quantity, and its level curves are invariant curves that orbits cannot leave. When both p and k are non-zero there is in general no conserved quantity.

A symmetry. Let $S = R_y(\pi)$, i.e. $(X, Y, Z) \mapsto (-X, Y, -Z)$. S commutes with $R_y(p)$. It flips $\hat{\boldsymbol{z}} \to -\hat{\boldsymbol{z}}$, so $SR_z(\alpha)S^{-1} = R_z(-\alpha)$, and it flips $Z \to -Z$, so the torsion angle changes sign as well. The two sign changes cancel:

$$F(S\boldsymbol{s}) = S\,F(\boldsymbol{s}). \tag{24}$$

The image of an orbit under S is therefore again an orbit. In the coordinates (ϕ, Z), S is the point reflection $(\phi, Z) \mapsto (\pi - \phi, -Z)$ about $(\pi/2, 0)$, and every phase portrait in these notes is symmetric under it. Orbits either come in symmetric pairs or are symmetric themselves.

6 Fixed points, periodic orbits and their stability

6.1 Linear stability of an area-preserving map

A point $\boldsymbol{s}^*$ with $F(\boldsymbol{s}^*) = \boldsymbol{s}^*$ is a *fixed point*. A point with $F^m(\boldsymbol{s}^*) = \boldsymbol{s}^*$ (and m minimal) lies on a *periodic orbit* of period m. To see what happens near $\boldsymbol{s}^*$, write $\boldsymbol{s} = \boldsymbol{s}^* + \delta\boldsymbol{s}$ with $\delta\boldsymbol{s}$ small and tangent to the sphere. To first order

$$\delta\boldsymbol{s}_{n+1} = M\,\delta\boldsymbol{s}_n, \tag{25}$$

where M is the 2×2 Jacobian of F at $\boldsymbol{s}^*$, written in any basis of the tangent plane (for a period-m orbit, M is the product of the m Jacobians along the orbit). Area preservation gives $\det M = 1$, so the eigenvalues satisfy $\lambda_1\lambda_2 = 1$ and solve $\lambda^2 - (\operatorname{Tr} M)\lambda + 1 = 0$. Everything is decided by the trace:

- $|\operatorname{Tr} M| < 2$: **elliptic** (stable). $\lambda_{1,2} = e^{\pm i\alpha}$ with $\cos\alpha = \operatorname{Tr} M/2$. Nearby points circle around $\boldsymbol{s}^*$ by an angle of about α per kick and stay close. In the phase portrait an elliptic point sits at the centre of a family of closed curves, an *island*.

- $|\operatorname{Tr} M| > 2$: **hyperbolic** (unstable). Real eigenvalues λ and $1/\lambda$ with $|\lambda| > 1$. Nearby points are pushed away along one direction and pulled in along another, and separations grow like $|\lambda|^n$. Hyperbolic points are the seeds of chaos.
- $|\operatorname{Tr} M| = 2$: **parabolic**, the boundary case at which **bifurcations** happen. At $\operatorname{Tr} M = +2$ ($\lambda = 1$) new fixed points can branch off. At $\operatorname{Tr} M = -2$ ($\lambda = -1$) a fixed point typically gives birth to a period-2 orbit (*period doubling*).

6.2 The poles $\pm\hat{\boldsymbol{y}}$

The two points $\boldsymbol{s} = \sigma\hat{\boldsymbol{y}}$, $\sigma = \pm 1$, are fixed for all p and k. They lie on the rotation axis of R_y, and they sit on the equator $Z = 0$, where the torsion angle vanishes. Near them use the tangent coordinates (x, z), $\boldsymbol{s} \approx (x, \sigma, z)$. The rotation (16) acts as the 2×2 rotation $\begin{pmatrix} c & s \\ -s & c \end{pmatrix}$ ($c = \cos p$, $s = \sin p$), and the torsion (18) to first order gives $x'' = x' - \sigma k z'$, $z'' = z'$. Hence

$$M_\sigma = \begin{pmatrix} 1 & -\sigma k \\ 0 & 1 \end{pmatrix} \begin{pmatrix} c & s \\ -s & c \end{pmatrix} = \begin{pmatrix} c + \sigma k s & s - \sigma k c \\ -s & c \end{pmatrix}, \qquad \det M_\sigma = 1, \tag{26}$$

$$\boxed{\operatorname{Tr} M_\sigma = 2\cos p + \sigma\, k \sin p.} \tag{27}$$

For $p = \pi/2$ this is $\operatorname{Tr} M_\pm = \pm k$. Both poles are stable for $k < 2$, and both become unstable at $k = 2$: $+\hat{\boldsymbol{y}}$ through $\operatorname{Tr} = +2$ and $-\hat{\boldsymbol{y}}$ through $\operatorname{Tr} = -2$. For small p, $\operatorname{Tr} M_+ \approx 2 - p^2 + kp$ exceeds 2 when $k > p$, the same threshold found for h_{eff} in section 4.

6.3 The case $p = \pi/2$

From now on we use the standard choice $p = \pi/2$. Then $R_y(\pi/2)$ maps $(X, Y, Z) \mapsto (Z, Y, -X)$, the torsion angle is $kZ' = -kX$, and the map (19) becomes

$$X_{n+1} = Z_n \cos(kX_n) + Y_n \sin(kX_n), \quad Y_{n+1} = -Z_n \sin(kX_n) + Y_n \cos(kX_n), \quad Z_{n+1} = -X_n. \tag{28}$$

The fixed points born at $k = 2$. A fixed point needs $Z = -X$, and then the first two equations of (28) both reduce (using $1 + \cos a = 2\cos^2\frac{a}{2}$ and $\sin a = 2\sin\frac{a}{2}\cos\frac{a}{2}$) to $Y = X\cot(kX/2)$. Normalisation $X^2 + Y^2 + Z^2 = 1$ then gives

$$2X^2 + X^2\cot^2\Big(\frac{kX}{2}\Big) = 1, \qquad \boldsymbol{s}_\pm = \big(\pm X,\ X\cot(kX/2),\ \mp X\big). \tag{29}$$

As $X \to 0$, $X\cot(kX/2) \to 2/k$, so a solution with small X requires $Y \approx 2/k \le 1$, i.e. $k \ge 2$. The pair $\boldsymbol{s}_\pm$ branches off the pole $+\hat{\boldsymbol{y}}$ exactly at $k = 2$, where its trace passes $+2$: a *pitchfork bifurcation*. (A numerical scan confirms that (29) has no solution for $k < 2$.) The two points are mapped onto each other by the symmetry S of (24). Solving (29) numerically and computing the trace of the Jacobian:

k	$\boldsymbol{s}_+ = (X, Y, Z)$	(ϕ, Z)	$\operatorname{Tr} M$
2.5	(0.514, 0.687, −0.514)	(0.928, −0.514)	1.00
3.0	(0.629, 0.456, −0.629)	(0.627, −0.629)	0.06
3.5	(0.681, 0.272, −0.681)	(0.380, −0.681)	−0.77
4.0	(0.702, 0.118, −0.702)	(0.167, −0.702)	−1.47

The trace decreases with k and reaches -2 at $k = \pi\sqrt{2} \approx 4.443$. This value can be checked exactly. At $k = \pi\sqrt{2}$, (29) is solved by $X = 1/\sqrt{2}$, $Y = 0$ (since $\cot(\pi/2) = 0$). Linearising (28) there in the tangent basis $\hat{\boldsymbol{u}} = \hat{\boldsymbol{y}}$, $\hat{\boldsymbol{w}} = (\hat{\boldsymbol{x}} + \hat{\boldsymbol{z}})/\sqrt{2}$ gives $M = \begin{pmatrix} -1 & -\pi/\sqrt{2} \\ 0 & -1 \end{pmatrix}$, so $\operatorname{Tr} M = -2$. Beyond $k \approx 4.44$ the fixed points $\boldsymbol{s}_\pm$ are unstable (period doubling), and their islands disappear from the phase portraits (fig. 5, $k = 4.5$).

A period-2 orbit born at $k = 2$. At $k = 2$ the pole $-\hat{\boldsymbol{y}}$ passes $\operatorname{Tr} = -2$ and emits a period-2 orbit. For $k = 3$ it consists of the two points $(\phi, Z) = (-0.627,\ 0.629)$ and $(-2.515,\ -0.629)$, with $\operatorname{Tr} M^{(2)} = -1.93$ (stable, close to its own period doubling).

A period-4 orbit for every k. The four points $\hat{\boldsymbol{x}} \to -\hat{\boldsymbol{z}} \to -\hat{\boldsymbol{x}} \to \hat{\boldsymbol{z}} \to \hat{\boldsymbol{x}}$ form a period-4 orbit for all k. $R_y(\pi/2)$ maps them cyclically into each other, and each image lies either on the z axis or on the equator, where the torsion does nothing. Its stability depends on k in a non-monotonic way:

k	0.5	1.5	2.5	3.0	4.5	6.0
$\operatorname{Tr} M^{(4)}$	1.98	0.68	−1.99	0.42	21.2	−1.94

It is stable at $k = 3$ and at $k = 6$: its islands are visible around $(\phi, Z) = (0, 0)$, $(\pm\pi, 0)$ and near the poles $Z = \pm 1$ in fig. 5. At $k = 4.5$ it is unstable.

7 From regular to chaotic motion

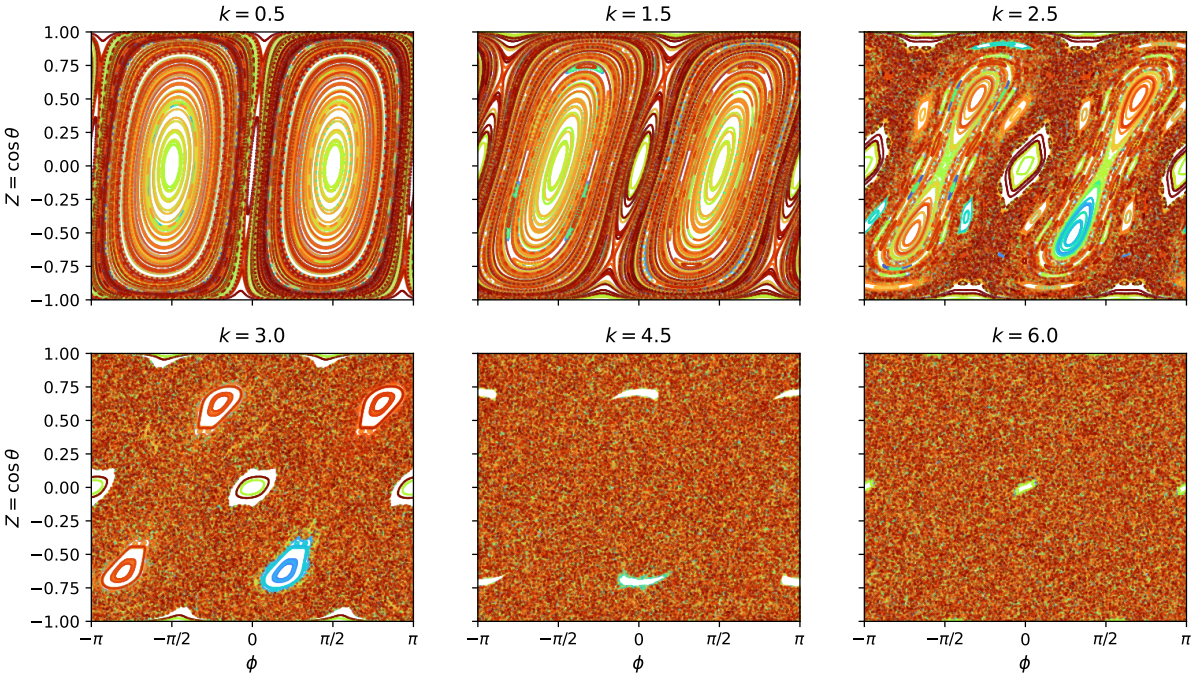


Figure 5: Phase portraits of the kicked top for $p = \pi/2$ and increasing k, in the canonical coordinates $(\phi, Z = \cos\theta)$ (equal areas in the plot are equal areas on the sphere). Each panel shows a 20×20 equal-area grid of initial conditions iterated for 500 kicks, one colour per orbit. Closed curves are regular (quasi-periodic) orbits. Uniformly speckled regions are filled by chaotic orbits.

7.1 Regular motion

In an integrable map every orbit lies on an invariant curve. If the motion along the curve is a rotation by an irrational fraction of a full turn per kick, the orbit fills the curve densely (*quasi-periodic* motion). If it is rational, the orbit is periodic. In a phase portrait, regular orbits appear as smooth closed curves.

For $p = \pi/2$ and small k (fig. 5, $k = 0.5$), the phase portrait looks almost integrable, although the unperturbed map $R_y(\pi/2)$ is degenerate (every orbit has period 4). The explanation is to look at four periods at once. Since $R_y(\pi/2)^4 = 1$, the map F^4 is close to the identity for small k, and to first order in k it is the flow of the torsion Hamiltonian $\frac{k}{2}Z^2$ averaged over the four rotated frames. Under successive rotations by $\pi/2$ about y, Z^2 becomes Z^2, X^2, Z^2, X^2, so

$$\bar{h} = \frac{k}{2}\left(Z^2 + X^2 + Z^2 + X^2\right) = k\,(1 - Y^2). \tag{30}$$

$\bar{h}$ depends only on Y, so Y is approximately conserved. The invariant curves are close to circles $Y = \text{const}$ around the poles $\pm\hat{\boldsymbol{y}}$. In the (ϕ, Z) plane these are the closed curves around $(\pm\pi/2, 0)$ in the $k = 0.5$ panel. On the great circle $Y = 0$ (the lines $\phi = 0, \pm\pi$) the averaged flow vanishes, so higher-order terms decide what happens there. Numerically a thin chaotic layer forms along these lines.

7.2 KAM theorem and the breakup of invariant curves

What happens to invariant curves when an integrable map is perturbed? The answer is given by the Kolmogorov–Arnold–Moser (KAM) theorem. Stated informally:

Theorem (KAM, informal)**.** *For a sufficiently small smooth perturbation of an integrable area-preserving twist map, most invariant curves survive, only slightly deformed. The survivors are those on which the rotation number is "sufficiently irrational". The destroyed curves occupy a fraction of phase space that goes to zero with the perturbation.*

The curves with rational rotation number r/s do not survive. By the Poincaré–Birkhoff theorem, such a curve breaks up into a chain of alternating elliptic and hyperbolic period-s points. Around the elliptic ones there are new small islands, and near the hyperbolic ones the stable and unstable directions intersect in an infinitely folded pattern (a *homoclinic tangle*), which produces a thin chaotic layer. The structure repeats on ever finer scales: islands around islands.

As k grows, the chaotic layers widen. When neighbouring resonances overlap (Chirikov's resonance-overlap criterion), the KAM curves that separated them are destroyed and the layers merge into a connected *chaotic sea*. In a 2D map, surviving KAM curves are impenetrable barriers: a chaotic orbit can never enter an island and a regular orbit can never leave one.

7.3 Mixed and global chaos in the kicked top

Figure 5 shows the scenario for $p = \pi/2$:

- $k = 0.5$ and 1.5: almost everything is regular. Large families of closed curves surround the stable poles $\pm\hat{\boldsymbol{y}}$ ($|\operatorname{Tr} M_\pm| = k < 2$), with thin chaotic layers near $\phi = 0, \pm\pi$.
- $k = 2.5$: the poles have just become unstable (section 6.2). Chaos spreads from them, and the new islands of $\boldsymbol{s}_\pm$ and of the period-2 orbit appear, together with a mixture of smaller island chains.
- $k = 3$: a *mixed phase space*. A large chaotic sea surrounds islands around $\boldsymbol{s}_\pm$ (at $(0.63, -0.63)$ and its mirror image $(2.51, 0.63)$), around the period-2 orbit (at $(-0.63, 0.63)$ and $(-2.51, -0.63)$), and around the period-4 orbit (at $(0, 0)$, $(\pm\pi, 0)$ and near $Z = \pm 1$). The pairing of islands under S of (24) is clearly visible.

- $k = 4.5$ and 6: almost *global chaos*. The islands of $\boldsymbol{s}_\pm$ are gone ($k > \pi\sqrt{2}$) and only tiny islands remain (e.g. the period-4 island at $(0, 0)$ for $k = 6$).

Figure 6 shows the difference between the two kinds of motion for $k = 3$. A single orbit started inside the island around $\boldsymbol{s}_+$ stays on a closed curve. A single orbit started next to the unstable pole $+\hat{\boldsymbol{y}}$ wanders through the whole chaotic sea but never enters any island. Figure 7 shows the $k = 3$ portrait on the sphere itself.

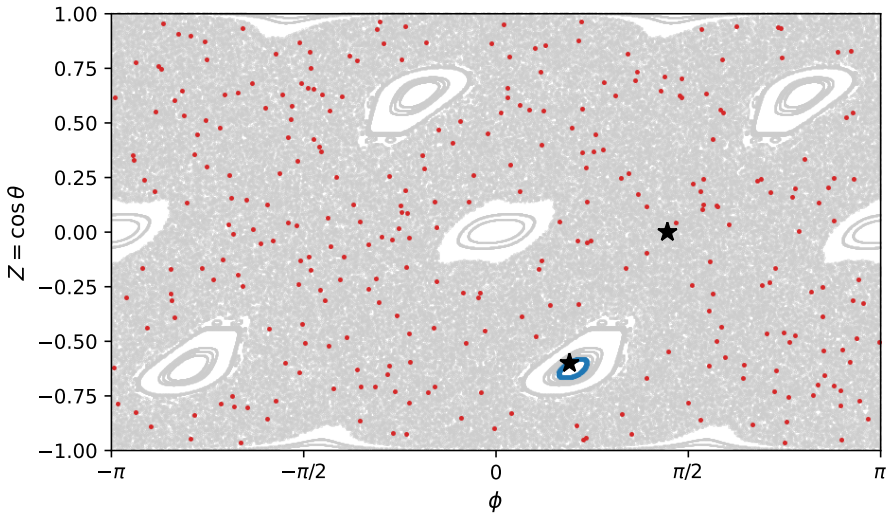


Figure 6: Two single orbits for $p = \pi/2$, $k = 3$, 300 kicks each, on top of the full portrait (grey). Blue: started at $(\phi_0, Z_0) = (0.6, -0.6)$, inside the island of $\boldsymbol{s}_+$. Red: started at $(1.4, 0)$, near the hyperbolic pole $+\hat{\boldsymbol{y}}$. Stars mark the starting points.

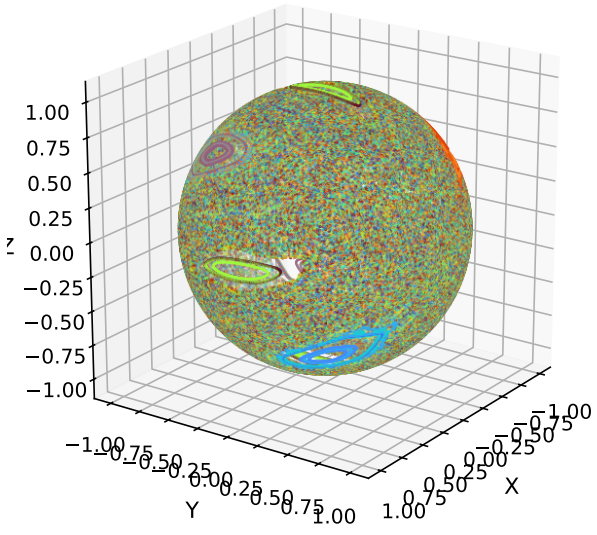


Figure 7: The $k = 3$ phase portrait of fig. 5 drawn on the unit sphere (X, Y, Z).

8 Lyapunov exponents

Chaos means *sensitive dependence on initial conditions*. Two orbits starting a small distance δ_0 apart separate exponentially, $\delta_n \sim \delta_0\, e^{\lambda n}$. The rate λ is the (maximal) *Lyapunov exponent*.

Formally, with the tangent map $M_n = DF(\boldsymbol{s}_{n-1}) \cdots DF(\boldsymbol{s}_0)$,

$$\lambda = \lim_{n\to\infty} \frac{1}{n} \ln \frac{\|M_n\, \delta\boldsymbol{s}_0\|}{\|\delta\boldsymbol{s}_0\|}, \tag{31}$$

which for almost every $\delta\boldsymbol{s}_0$ does not depend on the direction of $\delta\boldsymbol{s}_0$. For an area-preserving 2D map the two exponents are $\pm\lambda$, since areas are conserved. Typical values:

- regular (quasi-periodic) orbits: $\lambda = 0$. Nearby orbits on neighbouring invariant curves separate at most linearly in n, because their rotation numbers differ slightly;
- a hyperbolic fixed point with eigenvalue λ_u: the local rate is $\ln|\lambda_u|$. For the pole $+\hat{\boldsymbol{y}}$ at $k = 3$, $\operatorname{Tr} M = 3$ gives $\lambda_u = (3+\sqrt{5})/2$ and $\ln\lambda_u \approx 0.96$;
- chaotic orbits: $\lambda > 0$. The value is an average over the whole chaotic sea, which contains regions of very different local stretching.

Figure 8 shows this directly. Pairs of orbits are started with θ_0 differing by 10^{-8}. In the island the distance stays at $\sim 10^{-8}$. In the chaotic sea it grows exponentially until it saturates at the size of the sphere (at most 2). A straight-line fit of $\ln\delta_n$ for $5 \le n \le 40$ gives a slope of about 0.47 per kick. This is a rough, finite-time estimate from a single pair; a reliable value of λ needs (31) evaluated over long times with periodic renormalisation of the tangent vector (the method of Benettin *et al.*). It is not part of the code yet; section 11.5 shows where it would fit.

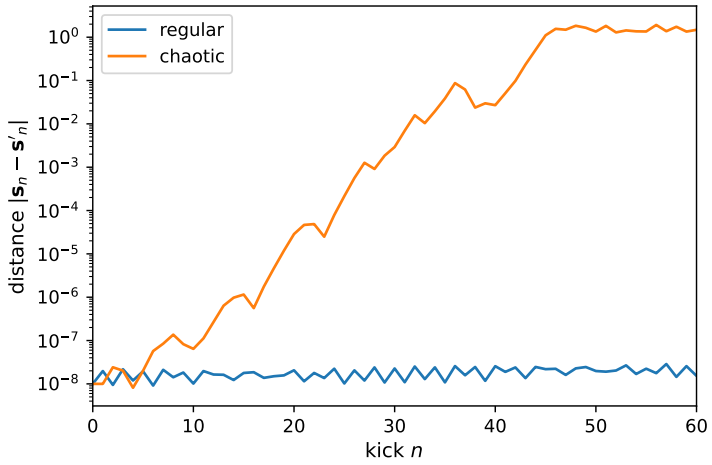


Figure 8: Distance between two orbits whose initial θ_0 differ by 10^{-8}, for $p = \pi/2$, $k = 3$. Blue: both inside the island of $\boldsymbol{s}_+$. Orange: both in the chaotic sea. The straight segment on the logarithmic axis is exponential separation.

9 Outlook: the quantum kicked top

The kicked top was introduced by Haake, Kuś and Scharf [1] as a model for *quantum* chaos. In quantum mechanics $\hat{\boldsymbol{J}}$ is a spin operator with $\hat{\boldsymbol{J}}^2 = j(j+1)$ (in units $\hbar = 1$), acting on a $(2j+1)$-dimensional space. The Hamiltonian (12) gives the Floquet operator over one period

$$\hat{U} = \exp\Big(-i\frac{k}{2j}\hat{J}_z^2\Big)\exp\big(-ip\hat{J}_y\big), \qquad |\psi_{n+1}\rangle = \hat{U}|\psi_n\rangle, \tag{32}$$

the quantum counterpart of (19). In terms of $\hat{\boldsymbol{s}} = \hat{\boldsymbol{J}}/j$ the commutator is $[\hat{s}_x, \hat{s}_y] = \frac{i}{j}\hat{s}_z$, so $1/j$ acts as an effective Planck constant and the classical map of these notes is the limit $j \to \infty$.

A spin coherent state, a minimum-uncertainty wave packet centred at a point $\boldsymbol{s}$ of the sphere, follows the classical orbit until it spreads. Spreading is slow on regular orbits, and on chaotic orbits it happens after a time of order $\ln j/\lambda$.

The classical phase-space structure leaves fingerprints in the quantum system. Examples include the statistics of the eigenphases of $\hat{U}$ (Poisson-like for regular, random-matrix-like for chaotic dynamics), the way eigenstates concentrate on classical islands or spread over the chaotic sea, and the entanglement generated when the spin is made of many qubits. The book by Haake [2] is the standard reference. The kicked top has been realised experimentally with the hyperfine spin of cesium atoms [8] and with superconducting qubits [9].

Part II
Numerics and code

10 Numerical method

Exact map, no integrator. The equations of motion (15) are solved exactly over one period (section 3), so the code never integrates a differential equation. It applies the closed-form map (19): two rotations per kick, 4 trigonometric evaluations and a few multiplications. There is no time-step error, and the only error is floating-point round-off. Since both steps are exact rotations, the norm $|\boldsymbol{s}| = 1$ and the area preservation of section 3.1 hold up to round-off. In practice $\big||\boldsymbol{s}_n| - 1\big| \lesssim 3 \times 10^{-15}$ after 500 kicks.

Round-off on chaotic orbits. On a chaotic orbit any error, including a round-off error of 10^{-16}, grows like $e^{\lambda n}$. With $\lambda \approx 0.5$ a computed chaotic trajectory differs from the exact trajectory with the same initial condition after roughly $\ln(10^{16})/0.5 \approx 70$ kicks. This does not make the phase portraits wrong. The computed orbit is still an orbit of the map to round-off accuracy at every step, and (by shadowing arguments) it stays close to some true orbit, so it explores the same chaotic sea. Individual chaotic trajectories beyond that time should be read statistically, not pointwise.

Initial conditions. Three modes are available.

- `single`: one point given by (θ_0, ϕ_0) through (13).
- `grid`: an $n_\theta \times n_\phi$ grid of cell centres, uniform in $Z = \cos\theta \in [-1, 1]$ and in $\phi \in [-\pi, \pi]$. By (9) every grid cell has the same phase-space area, so the grid samples all regions of phase space with equal weight. A grid uniform in θ would oversample the poles.
- `random`: points with Z and ϕ uniformly distributed, which by (9) is the uniform distribution on the sphere.

Recording. After `n_transient` discarded kicks, the state is stored at kicks $n = 0, m, 2m, \cdots \leq$ `n_kicks`, where $m =$ `save_every`. The initial point is stored as $n = 0$ and the map is applied once per kick, so stored points are always "just after a kick", as in section 3.

Parallelisation. Different initial conditions are independent, so the loop over them is parallelised with OpenMP. The default configuration (20×20 grid, 500 kicks) runs in under a second.

11 Code structure

The physics is in Fortran (`src/`); the Fortran code writes plot-ready files, and Python only reads and plots them. Figure 9 shows the data flow and table 1 maps each module to the equations it implements.

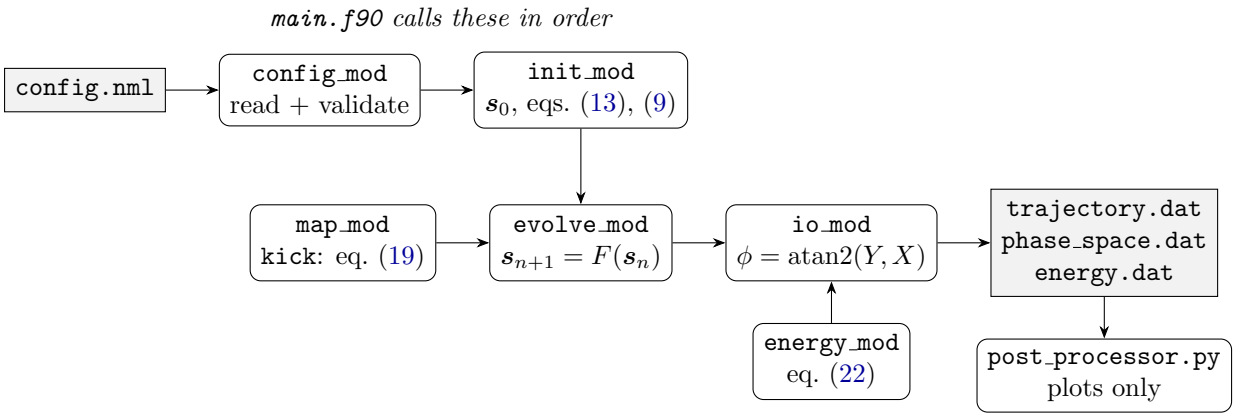


Figure 9: Data flow of the program. `main.f90` reads the configuration, builds the initial conditions, evolves them and writes the output.

File	Role	Physics
`kinds_mod.f90`	working precision `wp` (double), constants π, 2π	–
`config_mod.f90`	type `config_t`; reads the namelist groups, checks values	parameters p, k of (12)
`init_mod.f90`	`initial_conditions`: `single` / `grid` / `random`; `spherical_to_cartesian`	(13); equal-area grid from (9)
`map_mod.f90`	pure function `kick(s, p, k)`: one period	(16), (18), (19)
`evolve_mod.f90`	iterates `kick` for all initial conditions (OpenMP), transient and `save_every`	$s_n = F^n(s_0)$
`energy_mod.f90`	pure function `energy(s, p, k)` $\to [pY,\ \frac{k}{2}Z^2,\ \text{total}]$	(22)
`io_mod.f90`	writes the three ASCII files; computes $\phi = \mathrm{atan2}(Y, X) \in (-\pi, \pi]$	coordinates of section 1.3
`main.f90`	driver: `read_config` $\to$ `initial_conditions` $\to$ `evolve` $\to$ `write_results`	–
`post_processor.py` (in `scripts/`)	class `PostProcessor`: phase-space plot (ϕ, Z), 3D plot on the sphere	–

Table 1: Files of the code and the equations they implement.

The map. The function `kick` in `map_mod.f90` is a line-by-line transcription of (16) (lines 18–22 of the listing) and (18) (lines 24–28):

```
9   ! One period of the kicked top on the unit sphere (Haake convention):
10  ! rotation about y by p, then torsion about z by the angle k*Z.
11  pure function kick(s, p, k) result(s_new)
12    real(wp), intent(in) :: s(3)
13    real(wp), intent(in) :: p
14    real(wp), intent(in) :: k
15    real(wp) :: s_new(3)
16    real(wp) :: x, y, z, c, sn
17
```

```
    c = cos(p)
    sn = sin(p)
    x = s(1)*c + s(3)*sn
    y = s(2)
    z = -s(1)*sn + s(3)*c

    c = cos(k*z)
    sn = sin(k*z)
    s_new(1) = x*c - y*sn
    s_new(2) = x*sn + y*c
    s_new(3) = z
  end function kick
```

The equal-area grid. `grid_points` in `init_mod.f90` places cell centres uniformly in Z and ϕ, as required by (9):

```
    do i = 1, n_theta
      z = -1.0_wp + (real(i, wp) - 0.5_wp)*2.0_wp/real(n_theta, wp)
      do j = 1, n_phi
        phi = -pi + (real(j, wp) - 0.5_wp)*two_pi/real(n_phi, wp)
        ic = ic + 1
        s0(:, ic) = spherical_to_cartesian(acos(z), phi)
      end do
    end do
```

11.1 Parameters

All parameters are read from a namelist file (default `config.nml`; another file can be passed as the first command-line argument). Missing groups or variables keep their defaults.

Group	Variable	Meaning
`&system`	`p`, `k`	rotation angle p and torsion strength k in (12)
`&simulation`	`n_kicks`	number of kicks recorded after the transient
	`n_transient`	kicks discarded before recording
	`save_every`	store every `save_every`-th kick
`&initial`	`mode`	`'single'`, `'grid'` or `'random'`
	`theta0`, `phi0`	initial point for `single` (radians)
	`n_theta`, `n_phi`	grid size in $Z = \cos\theta$ and in ϕ
	`n_random`, `seed`	number of random points and seed
`&output`	`out_dir`	output directory

11.2 Output files

All files are ASCII, start with a header line `# p = ..., k = ...` and a column line, and contain one block per initial condition separated by a blank line. The three files have the same row order, so row r of each file refers to the same point $\boldsymbol{s}_n$.

File	Columns
`trajectory.dat`	orbit index `ic`, kick `n`, X, Y, Z
`phase_space.dat`	`ic`, `n`, ϕ, Z (canonical coordinates, section 1.3)
`energy.dat`	pY, $\frac{k}{2}Z^2$, total (eq. (22))

11.3 Building and running

```
make # build/kicked_top
./build/kicked_top config.nml # or: make run
python scripts/post_processor.py # output/phase_space.png
python scripts/post_processor.py --sphere --show
make figures # recompute the figures of these notes
make docs # compile docs/kicked_top_notes.tex
```

The figures of these notes are reproducible. `make figures` runs the program on every namelist in `docs/figures/configs/`, writing the output to `docs/figures/data/`. Then the script `plot_figures.py` in `docs/figures/` draws the PDF figures from those files.

11.4 Verification

The following checks were carried out on the code:

- $k = 0$, $p = \pi/2$: the map is $R_y(\pi/2)$, so every orbit must return to its starting point after 4 kicks. Observed error: 5×10^{-16}.
- $\big||\boldsymbol{s}_n| - 1\big| \le 3.4 \times 10^{-15}$ for the default run (400 orbits, 500 kicks, $k = 3$).
- The three columns of `energy.dat` agree with pY, $\frac{k}{2}Z^2$ and their sum recomputed from `trajectory.dat` to 10^{-15}.
- The fixed points, traces and symmetry of sections 5 and 6 were checked against the map numerically, and they agree with the islands in fig. 5.

11.5 Extending the code

New functionality follows the same pattern: one module with its own namelist group and an on/off switch, called from `main.f90`. For example, a Lyapunov exponent module would

1. add a pure function for the tangent map $DF(\boldsymbol{s})\,\delta\boldsymbol{s}$ (the derivative of (19)) next to `kick` in `map_mod`;
2. add `lyapunov_mod.f90`, which evolves $(\boldsymbol{s}, \delta\boldsymbol{s})$ together, renormalises $\delta\boldsymbol{s}$ every kick and accumulates $\ln\|\delta\boldsymbol{s}\|$ to estimate (31) for every initial condition;
3. add a group `&analysis lyapunov = .true. /` to the namelist and a file `lyapunov.dat` (`ic`, λ) written by `io_mod`, which Python can plot as a colour map over the initial conditions.

Other variants (a different kick axis, several kicks per period, dissipation) only replace `kick`. The rest of the program does not depend on the form of the map.

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